Nikhil Marudai

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Honestly, I do not know what to call myself, so here is a little of where I have been.

I studied human biology and bioinformatics in college. I spent four years in immunology research, on lung fibroblasts and influenza. I got into medical school and deferred it to build in AI, because AI can make a bigger impact. For the past six months I have been living on Claude Code, building and learning. Currently I am working on Organism, an AI co-scientist, and alongside it a social accountability app.

Everything you learn is a tool, and the more tools you have, the more problems you can solve, and the harder those problems can be, because hard problems rarely sit inside one field. You start seeing the same patterns in different fields, and a solution from one carries into another. I go learn whatever a problem needs from me, and not stay limited by what I know now.

SELECTED WORK

Organism - AI co-scientist for RNA sequencing data · genorganism.onrender.com · demo · 2026

Lets biologists analyze their own sequencing data in plain English, end to end, without writing code. An agent loops over hundreds of typed, provenance-tracked biology tools against a live analysis kernel, surfacing verification at every step so a scientist can check the reasoning rather than trust it blindly. In minutes it helped me find insights in my lab's data that I had not found in four years. Piloted by researchers in the Boyd Lab at UCSC.

REACH - social accountability platform · reachtoreach.com · App Store · 2025–2026

Shipped solo to the App Store. Users commit to a goal publicly, post daily photo or video proof, and real peers verify it, with streaks and seasons layered on top. Django backend, Capacitor mobile. Real users, with top streaks of 68 and 144 days.

More at nikhilmarudai.github.io/portfolio - over 4,300 commits since January 2026, including "intelligence," a personal AI memory substrate that I run on my own life, and hackathon builds.

RECOGNITION

- **re:AGENT Hackathon 2026** (co-hosted by Anthropic, Arc Institute, Biohub) - Won for groundloop, a verification layer for AI co-scientists that stops agents building on wrong claims on long trajectories.
- **Koret Scholars Research Award**, UC Santa Cruz - for single-cell RNA-seq analysis of fibroblast activation.
- **Y Combinator 2026** - application ranked in the top 10% of applicants.
- **npx standout** - ranked top 5% AI-native engineer.
- **GMI Cloud Hackathon 2026** - Won the Photon sponsor prize for Hoppy, an iMessage agent that turns a plain meal request into a store-ready grocery cart, using a memory of your preferences.

TECHNICAL

AI engineering - LLM agent architectures, tool harnesses, retrieval and memory systems, context engineering, evaluation harness design, provenance and verification layers.

Languages & tools - Python, R, bash, git

Bioinformatics - single-cell and bulk RNA-seq, trajectory analysis, cell-type annotation, differential expression, multi-dataset integration

Wet lab - developed protocols from scratch (immunofluorescence, RNA in situ hybridization for lung tissue), flow cytometry, tissue processing; the bench experience behind the bioinformatics

Product - web and mobile, shipped to the App Store

RESEARCH EXPERIENCE

Research Specialist - Boyd Lab, UC Santa Cruz · Jun 2023 – Dec 2025
How lung fibroblasts shape the immune response during influenza and other respiratory viral infections. Planned and led my own experiments. Combined wet-lab work with computational analysis: developed immunofluorescence and RNA in situ hybridization protocols for lung tissue, and built a bulk RNA-seq analysis pipeline. Ran intensive flow cytometry campaigns.

Undergraduate Researcher - Boyd Lab, UC Santa Cruz · Oct 2021 – Jun 2023
First person to join the lab as it was being set up. Before there was a wet lab, studied fibroblast activation during infection using single-cell RNA-seq. Received the Koret Scholars Research Award for this work, then taught labmates the tools and analysis.

PUBLICATIONS

1. Flerlage T, Boyd DF, Clark B, **Marudai N**, et al. *Integrated longitudinal transcriptomic and proteomic analysis of the murine lung response to influenza A virus.* **Am J Respir Cell Mol Biol** 2026;74(4):480–496. doi:10.1165/rcmb.2024-0405OC
2. Saini S, **Marudai N**, et al. *Lung adventitial fibroblasts direct compartmentalized inflammatory monocyte recruitment during influenza virus infection.* **In revision, Nature Immunology.**
3. **Marudai N**, Vaughn Jordan S, Savala D, et al. *Influenza A virus infection activates age-dependent inflammatory responses in lung fibroblasts.* **Submitted, Am J Respir Cell Mol Biol** (2026). *First author.*

PUBLISHED ABSTRACTS

- **Marudai N**, Koirala A, Savala D, Mora MA, Saini S, et al. *Age-Dependent Lung Fibroblast Composition and Responses to Influenza Infection.* **Am J Respir Crit Care Med** 2025;211(Suppl 1). **American Thoracic Society International Conference**, San Francisco. *First author.*
- Saini S, **Marudai N**, Trussell O, Mora MA, Boyd DF. *Pi16+ adventitial fibroblasts promote monocyte recruitment via CCL2-CCR2 signaling following influenza A virus infection.* **J Immunol** 2025;214(Suppl 1). doi:10.1093/jimmun/vkaf283.2118

- Mora MA, **Marudai N**, Boyd DF. *Complex Lung Organoids Model Fibroblast Responses to Influenza Infection.* **Am J Respir Crit Care Med** 2025;211(Suppl 1). **American Thoracic Society International Conference**.
- First-author posters at the **Stanford BayViro Conference** (2024) and the **Koret Scholars Research Symposium** (2023).

EDUCATION

B.S. Human Biology, minor in Bioinformatics - University of California, Santa Cruz · 2023
Summa cum laude. Koret Scholar.

Accepted to medical school - USF Morsani College of Medicine · 2026, deferred to 2027
Waitlisted at Mayo Clinic and Stanford. MCAT 521 (98th percentile).

BEFORE THIS

Medical Assistant, Action Urgent Care, San Jose · Oct 2024 – Nov 2025. Front and back office across eight clinic locations: triage, vitals, EKGs, venipuncture, laceration repair, insurance verification. Used Spanish with patients. Shadowed medical oncology at Zangmeister Cancer Center (2025) and family medicine at Sutter Health (2022–2023).

Supplemental Instruction Leader, UCSC Learning Support Services · Jan 2021 – Jun 2023. Led large-group physics tutoring, aimed at undoing "physics is not for me" by connecting equations to physical meaning. One student went from struggling to tutoring herself.

Volunteer Builder and Health Coordinator, Tiny Village Spirit, Richmond CA · Oct 2024 – Mar 2026. Built a tiny-home village for homeless youth. From no construction experience to leading volunteer crews on roofing and fencing. Empty land to seven tiny homes, two communal yurts, a fence and a 100-tree orchard. Coordinated healthcare, dental and job resources with local clinics for future residents.

Camp Counselor, Camp Kesem, UCSC chapter · 2022 – 2026. Summer camp for children whose parents have cancer. Stayed involved after graduating.

PERSONAL

I have played soccer my whole life. I have trained calisthenics for over ten years. I played club rugby at UC Santa Cruz.

Languages: English, Tamil (native), Spanish (working proficiency).

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